

Analysis of Algorithms (BDSF25M/A)

Date: 25-Feb-2026

Quiz 01

Time: 20 min

Roll No: _____ Name: _____

Honor Code: By signing below, I am declaring under oath that
I will solve the quiz by myself solely and will not
Take any help from neighboring fellows.

Signature

Grading	CLO-1	CLO-2	CLO-3	CLO-4
	/	/	/	/

1. Define Algorithm.
2. What are the properties of an algorithm.
3. What is the meaning of following: **Algorithms is technology?**
4. When is an algorithm said to be correct?

5. The Syracuse problem is a very simple problem of arithmetic that is still unsolved today. It can be stated as follows: the Syracuse sequence starts from any natural number, the next number of the sequence depends upon its previous as, if n^{th} number is odd, the $n+1^{\text{th}}$ number is $3n+1$ and if n^{th} number is even $n+1^{\text{th}}$ number is $n/2$. Write the recursive implementation of following function to print Syracuse sequence start from n . *Note: you may write cpp code or just algorithm.*
- void syracuse(int n)**

6. The recursive formula for 01 Knapsack problem is as follows:

$$B(n, W) = \begin{cases} 0 & \text{when } n \leq 0 \text{ or } W \leq 0 \\ B(n-1, W) & \text{when } n > 0 \text{ and } w[n] > W \\ \max(B(n-1, W), B(n-1, W - w[n]) + v[n]) & \text{when } n > 0 \text{ and } w[n] \leq W \end{cases}$$

where B is the maximum benefit from n item and knapsack capacity W . The arrays w and v contain the weights and the values correspond to n items in the shop. Write the following recursive function to return the maximum benefit for the specified 01 Knapsack problem. *Note: you may write cpp code or just algorithm.*

double knapsack01(int n, int W, double w[], double v[])